

Metropolitan scope amplifies measured gender disparities in stochastic urban accessibility: Evidence from Bogotá’s 2023 mobility survey

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Abstract

Urban accessibility estimates are commonly conditioned on administrative boundaries, potentially concealing inequalities generated by cross-border mobility. We introduce a stochastic accessibility framework based on first-passage times in an origin–destination Markov network and apply it to Bogotá’s regional household mobility survey. Weighted trips define population and sex-specific transition kernels to employment, education, and health destinations. A symmetric decomposition separates residential-distribution and mobility-kernel components, while shared household-cluster bootstrap perturbations propagate uncertainty across nested territorial scopes. Paired comparisons show larger female–male first-passage gaps in the metropolitan network than within Bogotá D.C. for all three destination purposes after simultaneous inference. A four-scope analysis offers an exploratory view of this boundary response; intermediate values are not strictly monotone for education and health. Metropolitan amplification is concentrated in transition dynamics rather than residential composition. The results show that territorial scope is part of the measured inequality estimand rather than a neutral preprocessing choice. The framework is descriptive and does not identify causal effects of municipal boundaries.

Keywords: stochastic accessibility; first-passage time; human mobility; gender inequality; metropolitan networks; Bogotá

1 Introduction

Accessibility links transport systems to the opportunities that people can reach and is therefore central to transport equity [Hansen, 1959, Pereira et al., 2017]. Standard place-based indicators commonly combine a fixed travel cost with a cumulative or gravity-based opportunity function. These measures are useful for planning, but they usually treat connectivity as deterministic and take the study boundary as given. In metropolitan systems, however, observed mobility repeatedly crosses administrative borders. An estimate restricted to the central municipality can therefore answer a materially different question from an estimate based on the functional urban region.

Gender is a particularly relevant dimension of this boundary problem. Large-scale mobility studies have documented systematic gender differences in travel diversity, trip purpose, mode use, and exposure to urban infrastructure [Macedo et al., 2020, 2022, Battiston et al., 2023]. For Colombia, previous analyses of Bogotá and Medellín show that these differences interact with socioeconomic status and the spatial distribution of destinations [Macedo et al., 2022]. Conventional potential-accessibility analysis has also documented strong income- and mode-related inequalities in the Bogotá city-region [Guzman et al., 2017]. Yet it remains unclear whether the magnitude and mechanism of a gender accessibility gap are stable when the mobility system is defined administratively rather than functionally.

Random walks provide a natural language for studying access in complex networks. First-passage time (FPT) measures the expected number of transitions required to reach a target set and has been used to characterize urban intelligibility and transport-constrained segregation [Blanchard and Volchenkov, 2008, Neira et al., 2024]. Related work uses random-walk coverage times to compare urban segregation across systems [Sousa and Nicosia, 2022], while recent mobility studies quantify administrative and physical barriers [Pintér and Lengyel, 2025] and mode-specific layers of social mixing [Yang et al., 2026]. These studies establish the value of stochastic and behavior-sensitive representations, but they do not resolve how a subgroup accessibility gap changes when the same observed metropolitan mobility system is truncated at an administrative boundary. Unlike a shortest path, FPT depends on the full transition structure and therefore reflects the range and concentration of observed movements. This feature is valuable for studying mobility inequality, but direct subgroup estimation is vulnerable to sparse origin rows and unstable transition probabilities.

This paper makes four contributions. First, it defines stochastic accessibility to purpose-specific opportunity sets using weighted mobility-survey flows rather than a geographic-adjacency or scheduled-route walk. Second, it regularizes group-specific transition rows toward a population kernel according to their effective sample size. Third, it separates the female–male accessibility gap into symmetric residential-distribution and mobility-kernel components. Fourth, it introduces a multiscale territorial response design: four nested systems are constructed from Bogotá D.C. to the complete Bogotá–Region network, and the same household bootstrap perturbation is propagated through every scope. The novelty lies in the estimand and inferential design—the boundary-induced response of a subgroup FPT gap—rather than in random walks alone. The empirical question is whether administrative truncation merely rescales accessibility or systematically attenuates measured gender inequality as observed metropolitan interactions are restored.

2 Data

2.1 Bogotá–Region Mobility Survey

We use the processed microdata and 2023 zoning files from the official Bogotá–Region Household Mobility Survey [Observatorio de Movilidad de Bogotá, 2025]. The public release contains linked household, person, trip, and trip-stage modules with survey expansion weights. The regional analytical sample retained 97,796 trips with valid household, origin, destination, and positive trip weight, and 64,373 persons whose residential unit belonged to the accepted network. The Bogotá D.C.-internal system retained 76,168 trips and 48,373 persons.

Spatial units are Urban Transport Analysis Units (UTAMs). The regional transition network contains 141 nodes. Bogotá D.C. contains 116 zoned UTAMs; the largest directed strongly connected component contains 114 nodes, 100% of the internal trip weight, and 99.78% of the eligible residential weight. The analysis uses the sex categories reported in the survey (women and men). Results should not be generalized to identities not represented by these categories.

Figure 1 shows the complete survey geography and the nested territorial scopes used in the multiscale analysis. External municipalities are classified according to the cumulative share of bidirectional survey-weighted flow that they exchange with Bogotá, using the full population before any subgroup comparison.

2.2 Purpose-specific opportunity sets

Trip purposes were harmonized into employment, education, and health. For each purpose and territorial scope, destination UTAMs were ranked by weighted arrivals after requiring at least five sampled trips and an effective sample size of at least three. The primary target set is the smallest ranked set reaching 50% of supported weighted arrivals (the *core-50* definition). The district target sets contain 24 employment, 24 education, and 17 health UTAMs. Because targets are reconstructed inside each scope, the scope contrast captures the complete change in the measured mobility system, including its opportunity geography.

3 Methods

3.1 Weighted mobility kernels

Let W_{ij} be the survey-weighted number of observed trips from UTAM i to UTAM j . The population transition kernel is

$$P_{ij}^{(0)} = \frac{W_{ij}}{\sum_k W_{ik}}. \quad (1)$$

For group g , let $\hat{P}^{(g)}$ be the corresponding empirical kernel and let

$$\text{ESS}_i^{(g)} = \frac{(\sum_{r \in i, g} w_r)^2}{\sum_{r \in i, g} w_r^2} \quad (2)$$

be the effective sample size of trips leaving origin i . Sparse rows are regularized toward the population kernel:

$$P_{ij}^{(g)} = \omega_i^{(g)} \hat{P}_{ij}^{(g)} + (1 - \omega_i^{(g)}) P_{ij}^{(0)}, \quad \omega_i^{(g)} = \frac{\text{ESS}_i^{(g)}}{\text{ESS}_i^{(g)} + \tau}, \quad (3)$$

with primary value $\tau = 20$. The Markov chain is a model of aggregate population flows. A transition sequence is not interpreted as an observed individual's itinerary.

3.2 First-passage accessibility

For target set A , let $T_A = \inf\{t \geq 0 : X_t \in A\}$. The vector of mean first-passage times $h^{(g,A)}$ is zero on A and satisfies

$$h_{-A}^{(g,A)} = \mathbf{1} + Q^{(g,A)} h_{-A}^{(g,A)}, \quad h_{-A}^{(g,A)} = (I - Q^{(g,A)})^{-1} \mathbf{1}, \quad (4)$$

where $Q^{(g,A)}$ is the transient submatrix of $P^{(g)}$. Let $\mu^{(g)}$ be the survey-weighted residential distribution for group g , conditioned on starting outside A . Group accessibility is summarized as

$$M^{(g,A)} = \mu^{(g)\top} h^{(g,A)}. \quad (5)$$

Larger values indicate more expected mobility transitions before reaching the opportunity set and hence lower stochastic accessibility.

3.3 Residence–kernel decomposition

For women (w) and men (m), the total gap is $G = M^{(w,A)} - M^{(m,A)}$. To distinguish a change in the gap from a general change in the FPT scale, we also report the descriptive normalized gap

$$G_{\text{rel}} = \frac{M^{(w,A)} - M^{(m,A)}}{\{M^{(w,A)} + M^{(m,A)}\}/2}. \quad (6)$$

Inference is based on the prespecified absolute gap G ; G_{rel} is used as a scale-comparability diagnostic.

We use a symmetric two-factor decomposition:

$$R = \frac{1}{2}(\mu^w - \mu^m)^\top (h^w + h^m), \quad (7)$$

$$K = \frac{1}{2}(\mu^w + \mu^m)^\top (h^w - h^m), \quad (8)$$

so that $G = R + K$. Here R is associated with the residential starting distribution and K with differences in transition dynamics. The decomposition is algebraic and descriptive; it is not a causal mediation analysis.

3.4 Multiscale territorial-scope response and inference

We construct four nested mobility systems. The first retains trips, residences, and targets internal to Bogotá D.C. External municipalities are then ranked by their total bidirectional survey-weighted flow with Bogotá, using the complete population and without reference to sex or outcomes. The functional 50% and functional 80% systems add the smallest ranked sets reaching at least 50% and 80% of this interface flow; the fourth system is the complete Bogotá–Region network. The realized interface shares are $q_s \in \{0, 0.564, 0.824, 1\}$. Within every scope, the largest directed strongly connected component, population kernel, residential distributions, and core-50 purpose targets are reconstructed.

For each purpose, the endpoint contrast is

$$\Delta_{\text{scope}} = G_{\text{region}} - G_{\text{district}}, \quad (9)$$

and the multiscale response is summarized by the ordinary least-squares slope β in

$$G_s = \alpha + \beta q_s + \varepsilon_s, \quad (10)$$

evaluated over the four prespecified realized shares. A positive β indicates that the measured gender gap tends to increase as observed metropolitan interface flow is incorporated. This slope summarizes the constructed scopes and is not a causal or continuous boundary effect.

Sampling uncertainty is estimated with a joint Poisson(1) household-cluster bootstrap. Each of 500 replicates assigns one multiplier to every household and applies it simultaneously to person and trip records in all four territorial scopes. Pointwise percentile intervals and max- t simultaneous intervals across the three destination purposes are reported separately for the total gap, residential component, and kernel component. Scope-specific target sets and population shrinkage kernels are reconstructed at the point-estimate stage and held fixed during bootstrap inference.

Two additional diagnostics probe the interpretation of the primary endpoint contrast. First, a time-threshold accessibility proxy uses pooled, survey-weighted mean reported duration for observed directed origin–destination edges, the shortest travel-time path to each supported core-50 destination UTAM, and pooled weighted destination arrivals as opportunity weights. For women and men

separately, we average the share of weighted destinations reachable within 45 or 60 minutes over their residential distributions, conditional on starting outside the target set. We define female disadvantage as male minus female reachable share. Paired regional-minus-district benchmark contrasts receive percentile intervals from 500 shared Poisson household resamples with fixed pooled travel-time surfaces and destinations. This benchmark describes revealed OD connectivity and does not contain a group-specific transition kernel.

Second, we recompute the sex-specific empirical transition rows using only trips that depart from each household’s residential UTAM, keeping the full population reference kernel, purpose targets, network components, and residential distributions fixed. These home-origin results are point-estimate sensitivities. Finally, to probe uncertainty from target selection, we perform 150 additional paired household resamples with and without rebuilding the core-50 destination sets within each replicate. Both variants retain the original network components and population reference kernel, and their percentile intervals are exploratory rather than substitutes for the primary simultaneous max- t intervals.

4 Results

4.1 Gender gaps within each territorial scope

In Bogotá–Region, the female-minus-male mean FPT gap was 0.999 steps for employment (simultaneous 95% confidence interval [0.632, 1.366]), 0.827 for education [0.451, 1.204], and 0.705 for health [0.252, 1.159]. The corresponding Bogotá D.C.-internal estimates were 0.479 [0.289, 0.669], 0.399 [0.173, 0.624], and 0.240 [-0.046, 0.527]. Thus, the sign is consistent across all point estimates, although the district health gap does not exclude zero after family-wise correction.

Table 1 reports the group-specific levels underlying these differences. After normalization by the two-group mean FPT, the employment gap increased from 12.4% within Bogotá D.C. to 21.7% in Bogotá–Region. The corresponding changes were from 10.0% to 17.5% for education and from 4.7% to 10.7% for health. The regional amplification therefore remains present on the relative scale and is not solely a consequence of higher overall FPT levels. These normalized quantities are descriptive diagnostics and are not assigned the confidence intervals of the absolute gaps.

The residential contribution was close to zero in every scope-purpose combination and all of its

Table 1: Group-specific mean FPT levels and female-minus-male gaps on absolute and normalized scales. Relative gaps divide the absolute gap by the two-group mean FPT.

Scope	Purpose	Men	Women	Absolute gap	Relative gap (%)
Bogotá D.C.	Employment	3.616	4.095	0.479	12.4
Bogotá-Region	Employment	4.111	5.110	0.999	21.7
Bogotá D.C.	Education	3.789	4.187	0.399	10.0
Bogotá-Region	Education	4.306	5.133	0.827	17.5
Bogotá D.C.	Health	5.045	5.285	0.240	4.7
Bogotá-Region	Health	6.257	6.963	0.705	10.7

simultaneous intervals included zero. In contrast, the mobility-kernel contribution closely reproduced the total gap. Figure 2 summarizes the scope-specific estimates.

4.2 Paired metropolitan amplification

The regional-minus-district gap was 0.520 steps for employment (simultaneous 95% confidence interval [0.178, 0.861]), 0.429 for education [0.070, 0.787], and 0.465 for health [0.087, 0.843]. Restricting the system to Bogotá D.C. attenuated the point estimates by 52.0%, 51.8%, and 65.9%, respectively. All three paired contrasts remained positive under simultaneous correction.

Kernel contrasts were likewise positive for employment (0.542 [0.217, 0.866]), education (0.451 [0.108, 0.794]), and health (0.495 [0.136, 0.853]). Residential contrasts were small and their intervals included zero. The increase in the measured gap at regional scale is therefore associated with transition dynamics rather than with a detectable change in the female-male residential composition contrast (Figure 3).

This conclusion was stable in point-estimate sensitivity analyses over $\tau \in \{0, 10, 20, 50\}$. The regional-minus-district total-gap contrast ranged from 0.472 to 0.557 steps for employment, from 0.380 to 0.468 for education, and from 0.423 to 0.497 for health. Kernel contrasts remained positive throughout, residential contrasts remained small, and all regional-minus-district relative-gap contrasts were positive. These calculations assess point-estimate sensitivity; the simultaneous intervals refer to the primary $\tau = 20$ specification (Additional file 1, Table S1).

The scope contrast also remained positive across all 12 effective combinations of target definition, residential conditioning, and purpose. Its range was 0.033–0.520 steps for employment, 0.034–0.429 for education, and 0.368–0.699 for health. Kernel and relative-gap contrasts were positive in

every combination. The wide ranges, particularly for employment and education, show that target breadth and residential conditioning affect magnitude even though they do not reverse the observed metropolitan amplification (Additional file 1, Table S10).

4.3 Multiscale boundary response

The four-scope analysis reveals a positive overall response to metropolitan expansion, but not strict stepwise monotonicity for every purpose. Employment gaps increased from 0.479 in Bogotá D.C. to 0.698 at functional 50%, 0.722 at functional 80%, and 0.999 in the complete region. Education gaps were 0.399, 0.335, 0.434, and 0.827, while health gaps were 0.240, 0.486, 0.434, and 0.705.

The estimated total-gap slope per unit of interface-flow coverage was 0.451 for employment (simultaneous 95% confidence interval [0.161, 0.741]), 0.320 for education [0.024, 0.616], and 0.391 for health [0.052, 0.730]. The corresponding kernel-component slopes were 0.469 [0.175, 0.762], 0.338 [0.035, 0.641], and 0.417 [0.078, 0.755]. Residential-component slope intervals all included zero. Bootstrap probabilities of a fully nondecreasing sequence across the four scopes were 0.756 for employment, 0.110 for education, and 0.212 for health. Accordingly, the supported claim is a positive average boundary response across prespecified scopes, not universal local monotonicity. Figures 4 and 5 show the total and decomposed responses.

4.4 Municipal contributions

Among external municipalities, Facatativá had the largest positive kernel contribution for all three opportunity classes: 0.223 steps for employment, 0.308 for education, and 0.185 for health. Soacha contributed 0.204 for employment and 0.172 for health, while its education contribution was slightly negative. These are additive origin-level attributions under the fitted regional model, not effects of deleting municipalities. In particular, leave-one-municipality-out residential calculations retain the regional kernel and must not be interpreted as infrastructure interventions. Figure 6 displays the purpose-specific external-municipality attributions.

A separate full-refit diagnostic excluded each of the 20 external municipalities in turn and reconstructed the regional network, both sex-specific kernels, residential distributions, and purpose-specific targets. The regional-minus-district gap remained positive after every omission: 0.277–0.565 steps for employment, 0.080–0.564 for education, and 0.270–0.552 for health. Excluding Facatativá

yielded the smallest contrast for each purpose, with 0.277, 0.080, and 0.270 steps, respectively; it had 229 sampled outbound households. Excluding Soacha yielded 0.389, 0.560, and 0.347 steps, respectively, with 1,449 sampled outbound households. The regional-minus-district normalized-gap contrast also remained positive after every omission. These are point-estimate influence diagnostics: the confidence intervals for the primary analysis cannot be transferred to the refitted networks (Additional file 1, Table S11).

4.5 Comparison with time-based accessibility and home-origin kernels

The deterministic time-threshold proxy showed regional-minus-district female-disadvantage contrasts between -0.0008 and -0.0028 in weighted reachable-opportunity share across the three purposes and two thresholds. Five of six descriptive bootstrap percentile intervals included zero; the education contrast at 45 minutes was -0.0028 [-0.0054, -0.0001]. These shares cannot be compared numerically with FPT transitions. In this pooled-time benchmark, sex differences arise from residential starting distributions alone, whereas the FPT analysis also incorporates sex-specific transition patterns. The distinct boundary responses indicate that the two operational measures summarize different features of the observed system, without establishing that one is a more accurate measure of real opportunities (Additional file 1, Table S12).

When empirical female and male kernel rows were restricted to home-origin trips, the regional-minus-district FPT contrasts remained positive at 0.558, 0.504, and 0.333 transitions for employment, education, and health, respectively, compared with 0.520, 0.429, and 0.465 in the primary model. Home-origin trips represented 62.3% of pooled survey-weighted regional trips. This point-estimate diagnostic shows that non-home departures alone do not generate the sign of metropolitan amplification; it does not remove all trip chaining or identify its behavioral cause (Additional file 1, Table S12).

In the reduced paired bootstrap, reselecting destination sets yielded exploratory percentile intervals of [0.184, 0.799] for employment, [0.050, 0.904] for education, and [0.048, 0.928] for health. All remained above zero, but the education and health intervals widened relative to fixed targets while the employment interval narrowed. These 150-replicate intervals are unadjusted across purposes and do not replace the primary 500-replicate simultaneous intervals (Additional file 1, Table S13). Figure 7 displays these three distinct diagnostics without equating their units or interval definitions.

5 Discussion

The results support a simple but consequential empirical conclusion: an administrative definition of Bogotá captures substantially less measured gender inequality than the functional regional mobility system. This extends prior evidence on accessibility inequities in the Bogotá city-region [Guzman et al., 2017] by showing that territorial scope changes the estimated gender gap itself, not only aggregate accessibility levels. It also complements research on mobility barriers and multimodal segregation [Pintér and Lengyel, 2025, Yang et al., 2026]: here the boundary is treated as a paired change in the stochastic-accessibility estimand, with uncertainty propagated from the same sampled households in both scopes. The paired design shows that the female–male gap is attenuated by approximately one-half for employment and education and by roughly two-thirds for health when the analysis is restricted to Bogotá D.C. As a descriptive scale check, the relative gaps also increase from the district to the regional system for all three purposes, so the endpoint result is not explained solely by an increase in the overall FPT level. The four-scope fitted slopes are positive after family-wise correction but reuse the endpoint data and therefore provide a descriptive view of the constructed scopes rather than independent confirmation of the paired result. At the same time, the intermediate education and health estimates show that metropolitan expansion need not produce a locally monotone sequence, so the result concerns the overall response across prespecified nested scopes rather than a universal law of boundary growth.

The municipality omission diagnostic indicates that no single external municipality reverses the positive endpoint contrast at the point-estimate level. Facatativá has the largest downward influence on this contrast, particularly for education. This influence reflects a simultaneous change in network flows, sampled residences, and destinations, and its ranking need not match fixed-kernel origin attributions. The stability of the paired contrast across the shrinkage range indicates that metropolitan amplification is not an artifact of the primary regularization value. The magnitude decreases as stronger pooling brings the sex-specific kernels closer to the population kernel, but the sign and component-level interpretation are unchanged.

The decomposition clarifies the mechanism represented by the model. Women and men have similar weighted residential starting distributions after conditioning on each target set, but their estimated transition kernels generate different first-passage times. Regional peripheral origins

contribute disproportionately to this kernel contrast despite representing smaller residential masses. This pattern is compatible with gendered differences in trip purpose, mobility diversity, mode use, and constraints documented in Colombia and other urban systems [Macedo et al., 2020, 2022, Battiston et al., 2023]. The present analysis adds that these disparities may be understated when the network is truncated at the central city’s boundary.

The time-based comparator yields near-zero or negative changes in the residential access-share disparity as the boundary expands. This is consistent with the small residential contribution of the FPT decomposition and illustrates what additional variation a behavioral transition kernel can represent in this dataset. Because the comparator uses observed OD edges and destinations rather than a comprehensive opportunity inventory or calibrated transport-time network, the contrast cannot establish general superiority over conventional accessibility models. The home-origin check preserves positive FPT boundary contrasts, reducing the concern that non-home departures alone generate the result, although trip chaining remains a plausible part of the observed transition dynamics. Exploratory target reselection suggests that fixing purpose targets can narrow intervals for education and health; the original simultaneous intervals should be read as conditional on the primary target sets.

Methodologically, the framework complements deterministic accessibility measures. It does not ask for the shortest or fastest route from a residence to a facility. Instead, it asks how many transitions are expected before aggregate observed mobility reaches a set of purpose-relevant destinations. The measure is therefore sensitive to network concentration, indirect flow structure, and the accessibility of a class of opportunities rather than a single facility. This interpretation should be maintained in policy use: FPT steps are dimensionless mobility transitions, not minutes of individual travel.

Several limitations bound the conclusions. First, the analysis is cross-sectional and descriptive; neither the scope contrast nor the decomposition identifies causal effects. Second, observed trip destinations are revealed opportunities, not a census of jobs, schools, or health facilities. Third, group kernels aggregate heterogeneous modes, schedules, trip chains, and unobserved constraints. Fourth, survey weights address the sampling design but cannot eliminate recall error or nonresponse bias. Fifth, the binary sex categories available in the processed modules do not represent the full diversity of gender identities. Finally, scope-specific target sets mean that the scope response intentionally captures both network truncation and the associated change in opportunity geography.

Target breadth and residential conditioning materially change the estimated magnitude, especially for employment and education, although all paired point contrasts retain the same sign. The full municipal refits have no household-bootstrap uncertainty intervals, and municipalities differ markedly in sample support; their point estimates should not be interpreted as intervention effects. The four interface shares are constructed from one survey and only two intermediate thresholds; consequently, the fitted slope is a parsimonious summary rather than evidence of linearity over arbitrary boundaries. The conventional comparison is based on reported durations for observed OD edges and revealed destinations, without a complete multimodal travel-time surface or an independently measured inventory of opportunities. Fixed-target simultaneous bootstrap intervals are conditional on destination selection and the full-sample population reference; the reduced target-reselection bootstrap assesses only part of that uncertainty. Supplementary analyses with extended targets and alternative conditioning rules assess specification sensitivity (Additional file 1).

For urban policy, the findings imply that equity assessment should align its spatial unit with the functional mobility system. Bogotá’s access to employment, education, and health is jointly produced with surrounding municipalities. Evaluations confined to district boundaries can miss the transition patterns through which regional residents encounter opportunities. This does not prescribe a particular intervention, but it identifies where more detailed modal, temporal, and institutional analyses should focus.

6 Conclusions

Using a regularized first-passage framework and a shared household bootstrap, we find that measured female–male stochastic-accessibility gaps increase on average as the mobility system expands from Bogotá D.C. through nested functional scopes to Bogotá–Region. Positive multiscale slopes remain after simultaneous inference for employment, education, and health, although education and health are not strictly monotone at every intermediate boundary. The response is concentrated in the mobility-kernel component rather than the residential-start component. Territorial scope is therefore part of the estimand, not a neutral preprocessing choice. Treating the metropolitan mobility network as the primary system, while reporting the complete boundary-response curve, provides a more faithful and transparent account of regional accessibility inequality.

List of abbreviations

EODH: Household Origin–Destination Survey; ESS: effective sample size; FPT: first-passage time; UTAM: Urban Transport Analysis Unit.

Additional file

Additional file 1 (.pdf): Supplementary material. Regularization sensitivity, the complete robustness grid, regional, paired, and multiscale bootstrap inference, district-network diagnostics, socioeconomic-stratum results, external-municipality attribution, full municipal omission sensitivity, and time-access, home-origin, and target-selection checks.

Declarations

Availability of data and materials

The survey microdata and zoning files are publicly available from the Bogotá Mobility Observatory [Observatorio de Movilidad de Bogotá, 2025]. Reproducible analysis code is maintained at <https://github.com/ardominguezm/colombia-stochastic-access>. Before submission, the repository will be archived in a DOI-issuing repository and the persistent identifier inserted here. Survey microdata are not redistributed by the authors.

Competing interests

The author declares no competing interests.

Funding

Not applicable. [Confirm before submission.]

Authors' contributions

ARDM conceived the study, developed the methodology and software, performed the analysis, interpreted the results, and drafted the manuscript. [Update if co-authors are added.]

Acknowledgements

Not applicable. [Update before submission.]

Use of generative artificial intelligence

Generative artificial intelligence tools were used to assist with code review and English-language drafting. The author defined the research questions and analytical decisions, verified all computations and sources, and assumes full responsibility for the manuscript. This disclosure should be revised to match the journal’s policy and the final workflow at submission.

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Study area and nested territorial scopes

Bogotá–Region Household Mobility Survey, 2023

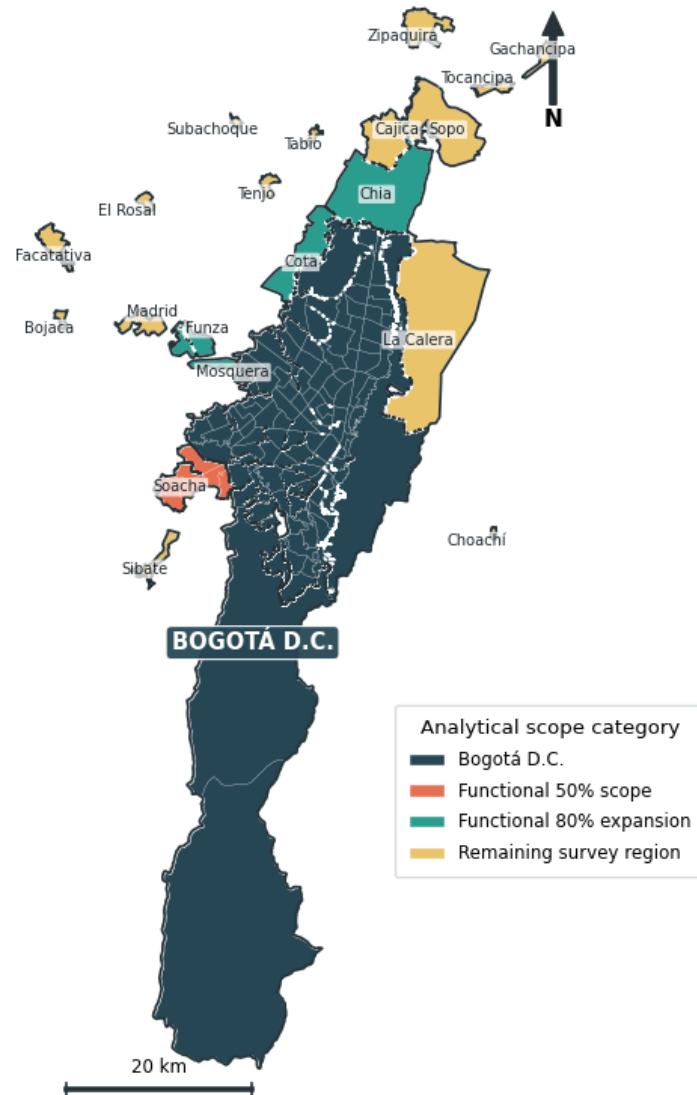


Figure 1: Study area and nested territorial scopes in the Bogotá–Region Household Mobility Survey. Bogotá D.C. forms the administrative core; Soacha reaches the functional 50% threshold; Mosquera, Chía, Funza, and Cota extend coverage beyond 80% of observed Bogotá–external interface flow; the remaining municipalities complete the regional system. Thin internal lines show UTAM boundaries.

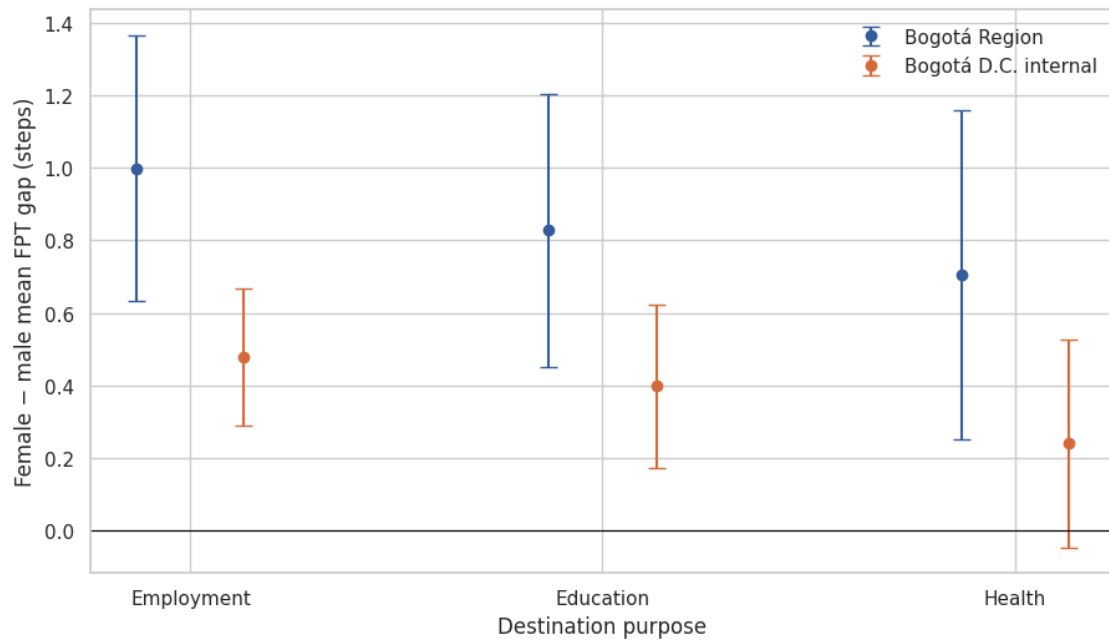


Figure 2: Gender gaps in stochastic accessibility by territorial scope

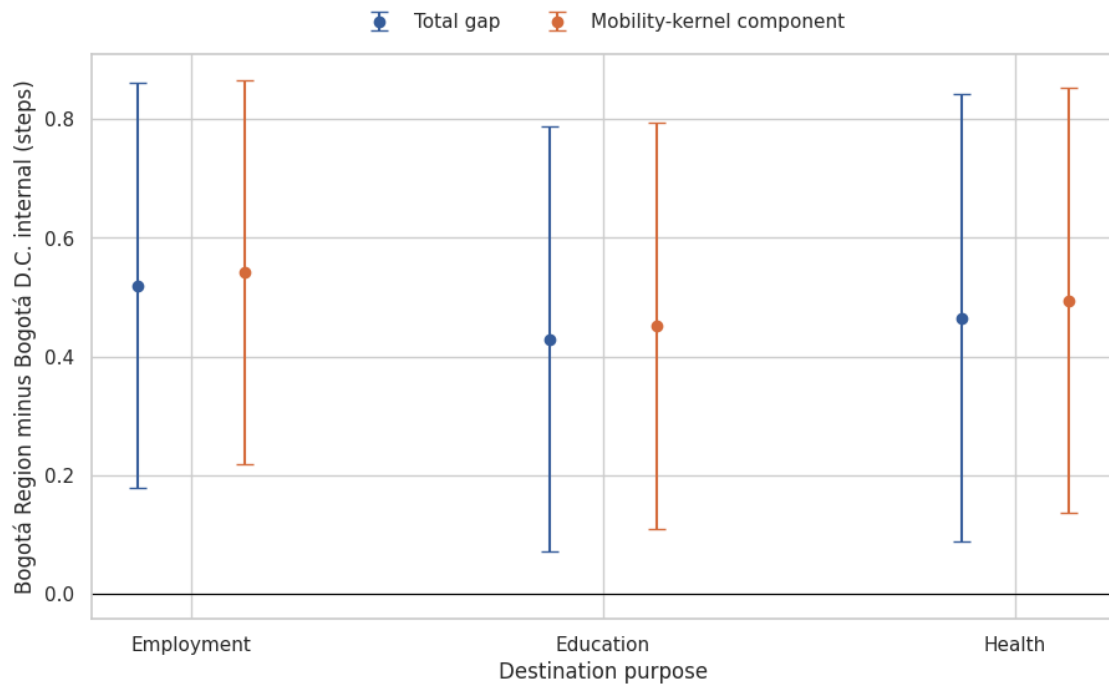


Figure 3: Paired amplification of gender gaps at regional scale

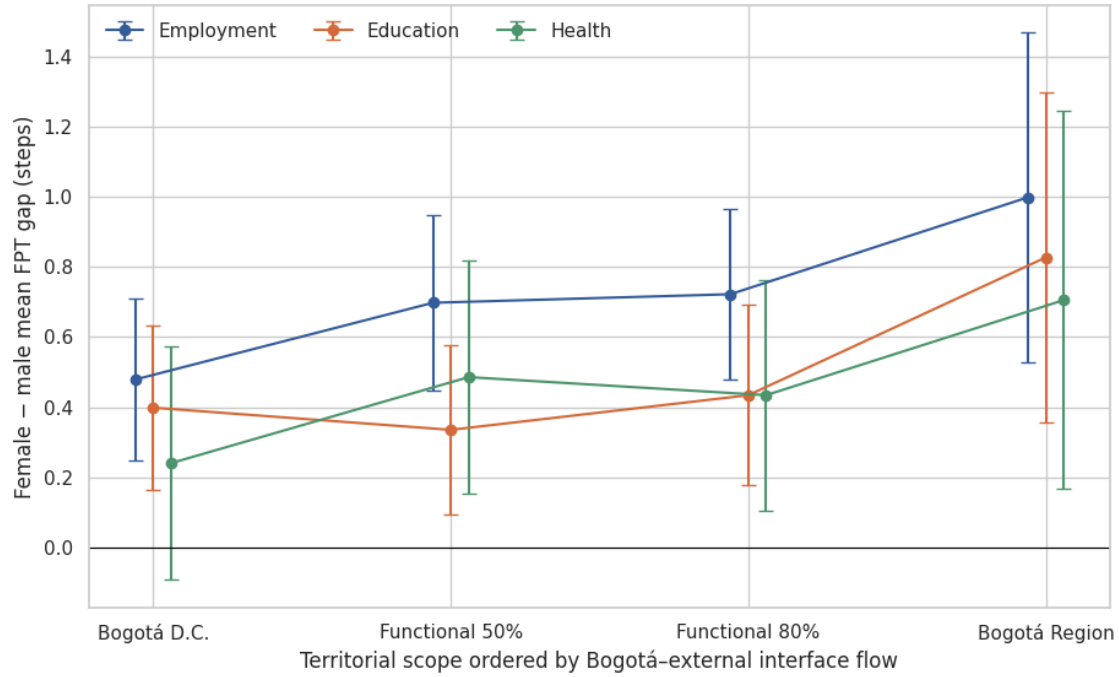


Figure 4: Multiscale response of the female–male stochastic-accessibility gap to progressively broader functional territorial scopes. Points show scope estimates; uncertainty and fitted slopes are obtained from the shared household-cluster bootstrap.

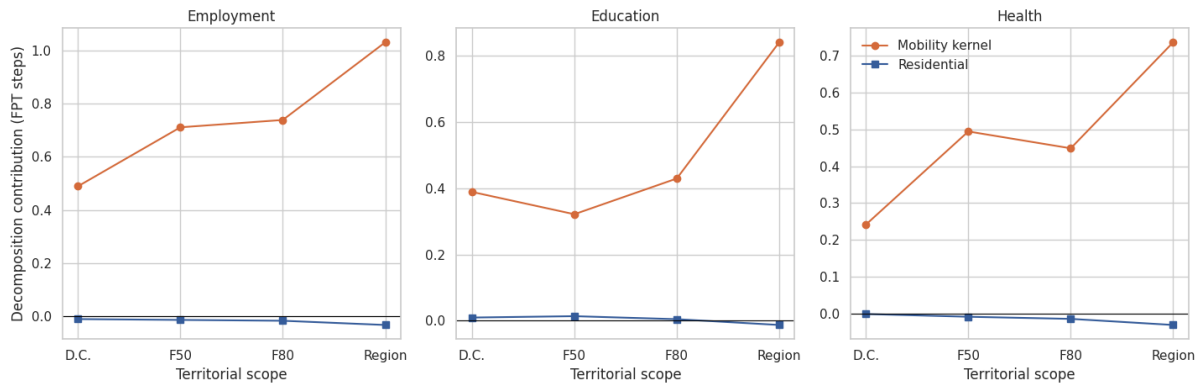


Figure 5: Multiscale decomposition of the territorial boundary response

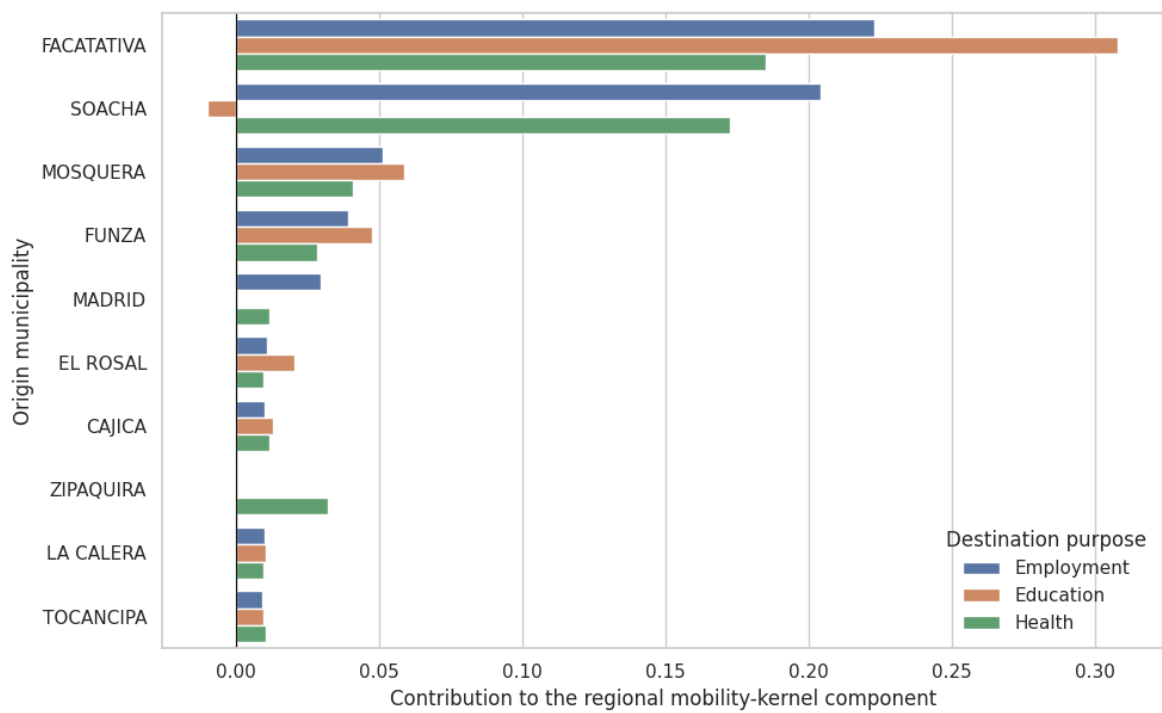


Figure 6: External-municipality contributions to the regional mobility-kernel component

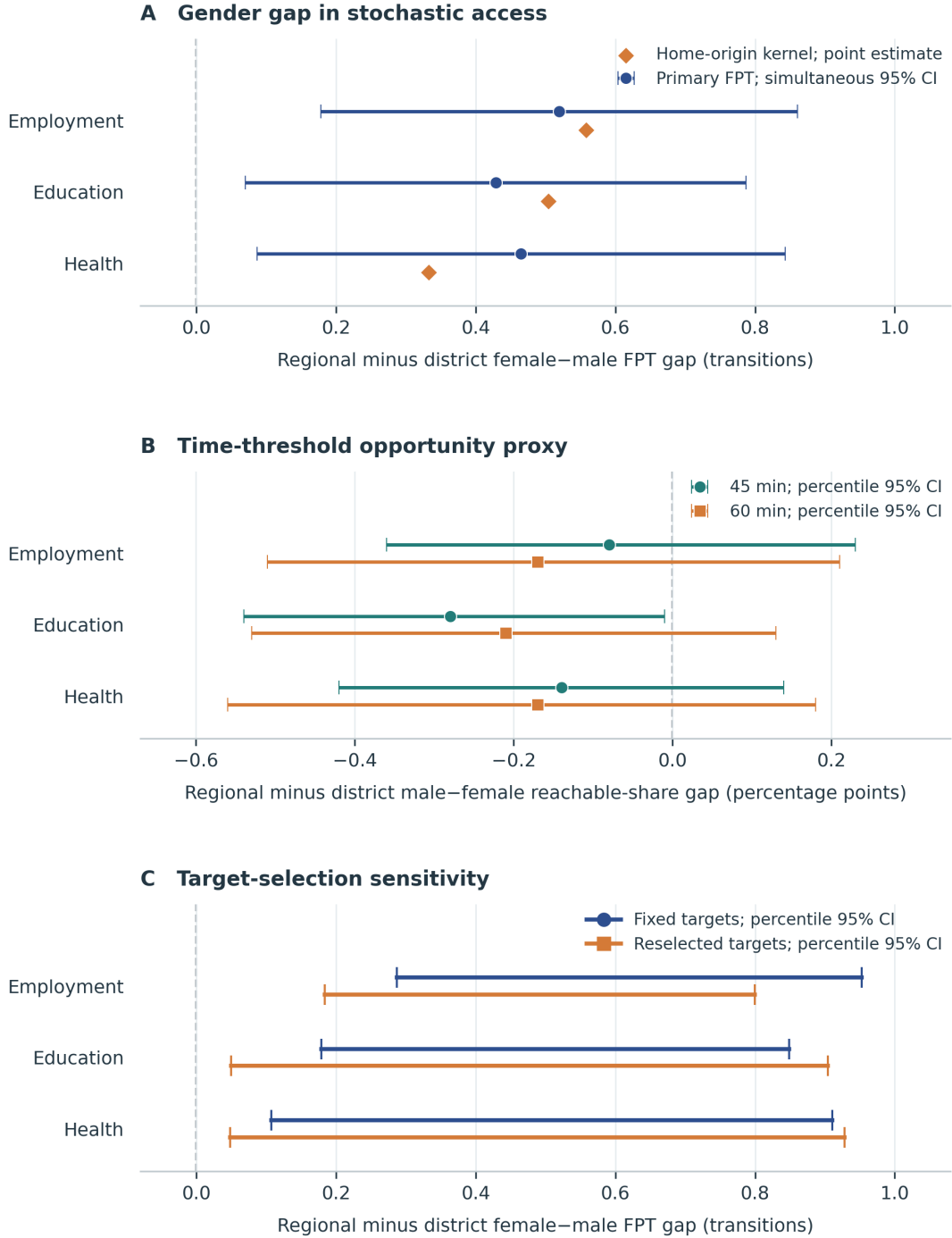


Figure 7: Sensitivity diagnostics for the regional gender gap in stochastic access. (A) Primary regional-minus-district first-passage-time (FPT) contrast and its simultaneous 95% confidence interval, alongside a home-origin-only point estimate. (B) Regional-minus-district contrast in the weighted share of destinations reachable within 45 or 60 minutes; conditional percentile intervals are descriptive and unadjusted. (C) Exploratory 150-replicate conditional percentile intervals for fixed and reselected opportunity sets. Each panel has its own units; panel C does not replace the primary inference in panel A.